Gravitational Waves

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M.Sc. 3rd Semester

Contents

- What are Gravitational Waves
- Formation
- First detection
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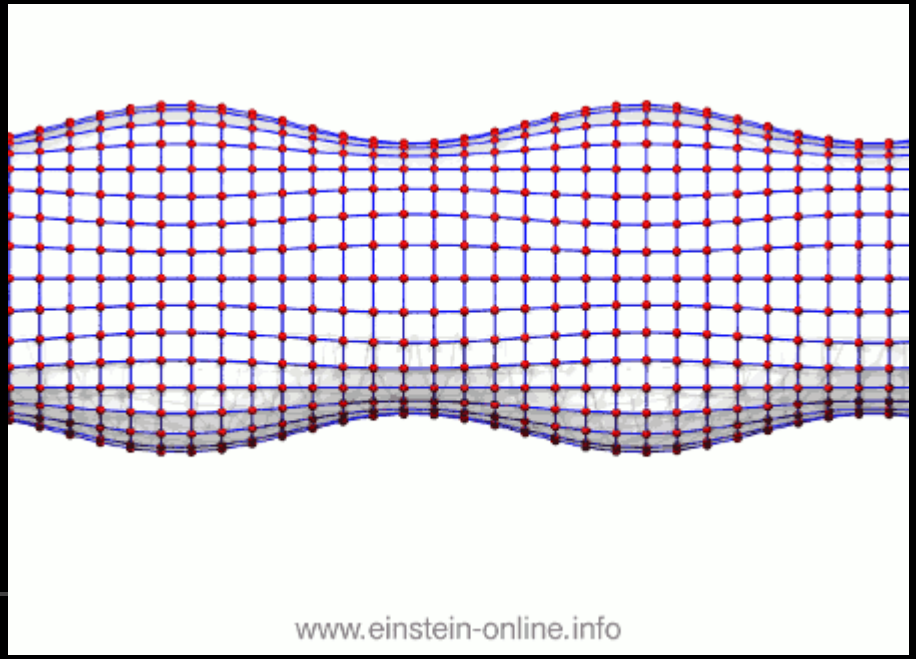
What are Gravitational waves

- Theoretically predicted by Einstein.
- Disturbance in Space-Time Curvature.
- Speed



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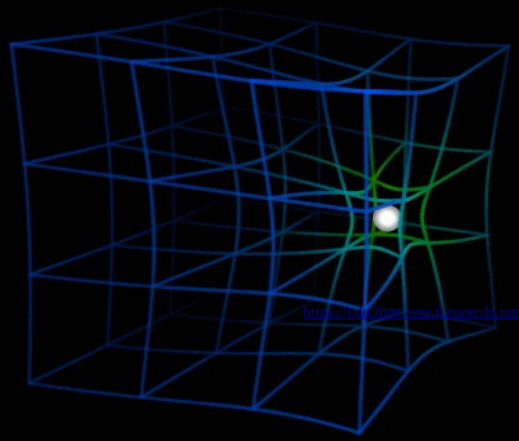
Formation



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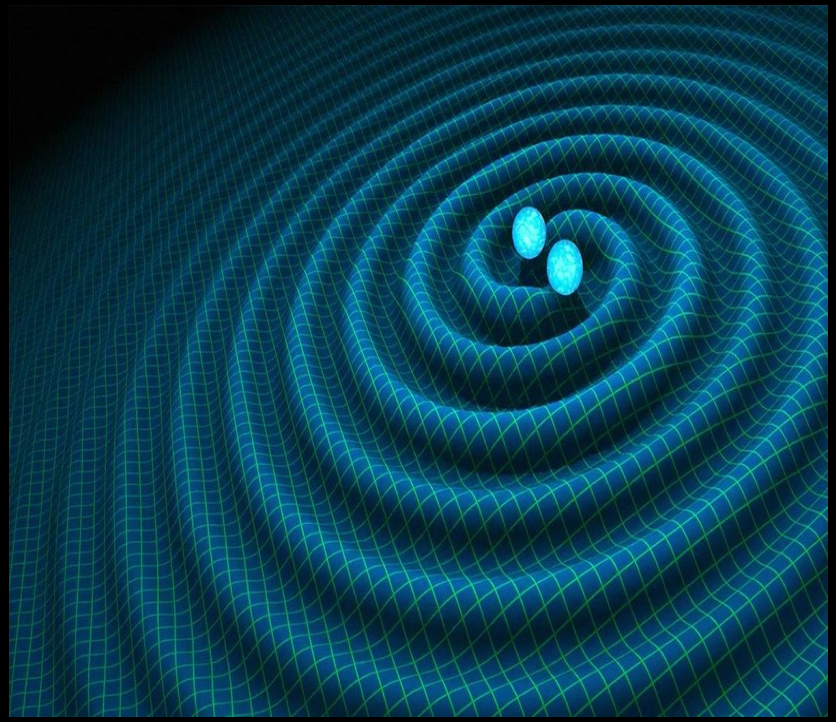

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Image source :-
https://cdn.mos.cms.futurecdn.net/ydwsrwtqwfCre2v6KPpJcn-1200-80.jpg

First Detection

- On September 14, 2015.
- By the Laser Interferometer Gravitational-Wave Observatory (LIGO).
- Gravitational waves generated by the collision of two Black Holes.
- The length of the arm got slightly changed.

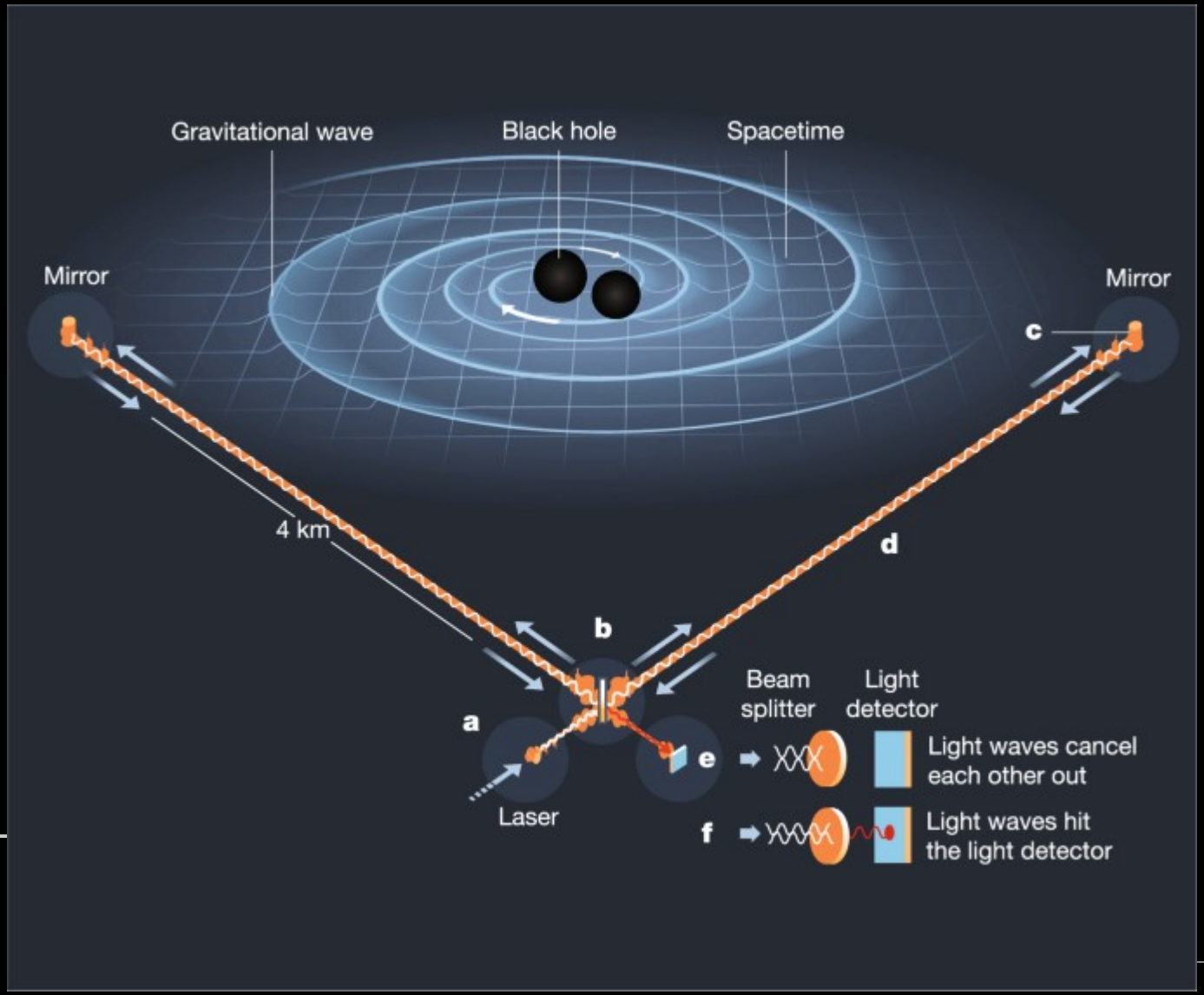

Image credit :-

https://media.springernature.com/m685/springer-static/image/art%3A10.1038%2Fs41586-019-1129-z/MediaObjects/41586_2019_1129_Fig1_HTML.png

Image credit :-- https://preview.redd.it/ligo-hanford-the-laser-interferometer-gravitational-wave-v0-i7fhgffbqvc1.jpeg?auto=webp&s=c64fd9cde618b99891fd6a3aec001dd01f8bc220

Why are they difficult to detect

- Incredibly Small Disturbance in space time.
- Noise from Other Sources.
- Sensitivity of Equipment.
- Precision of Timing and Synchronization.
- Detection at Great Distances.

Significance of detection

- Confirmation of Einstein’s Theory
- Opens New Window into the Universe
- Start of a new branch, gravitational wave astronomy.
- Led to Nobel Prize in Physics:-

  In 2017, the Nobel Prize in Physics was awarded to Rainer Weiss, Barry C. Barish, and Kip S. Thorne, the key scientists behind the LIGO project, for their decisive contributions to the detection of gravitational waves.

References

- Lydia Bieri, David Garfinkle, Nicol´as Yunes, Vol. 64, Issue 07, 2017, (August issue 2017) https://arxiv.org/pdf/1710.03272
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- "The Nobel Prize in Physics 2017: LIGO and the discovery of gravitational waves." https://www.nobelprize.org/prizes/physics/2017/press-release/
- B.P. Abbott et al., "Observation of Gravitational Waves from a Binary Black Hole Merger," *Physical Review Letters*, vol. 116, no. 6, 061102 (2016). DOI: 10.1103/PhysRevLett.116.061102 https://arxiv.org/abs/1602.03837

THANK YOU